Date ____ / /

Assignment

(1) Given
Lan bandwidth = 10 Mbps
Measured = 1200 frame/min
Total bits per minute = 12,000 x 10000
= 120000000 bits
Throught in bps = 120 000 000 / 60
= 2000000 bps
= 2 Mbps

(i) Why lower than 10Mbps
↳ Protocol overhead ; collisons / retransmission
↳ Application or host processing limits
↳ congestion or queuing delays
↳ Errors and retransmission lower effective throughput.

(2) Intrupt UDP header hex CB84000D001C001C

↳ Hex bytes (pairs):
CB 84 00 0D 00 1C 00 1C

UDP header fields

Source port = CB84 = 52100 (decimal)
Destination port = 000D = 13
Length (UDP total length) = 001C = 28 bytes
Checksum = 001C = 0x001C decimal 28)

Size of UDP data

UDP header = 8 bytes → Data size = 28 - 8
= 20 bytes.

↳ Ports are 16 bit fields in network byte order (big-endian).
↳ Length is the entire UDP datagram length.

(Q3) IP block 17.63.110.114/24
- first / last usable and broadcast

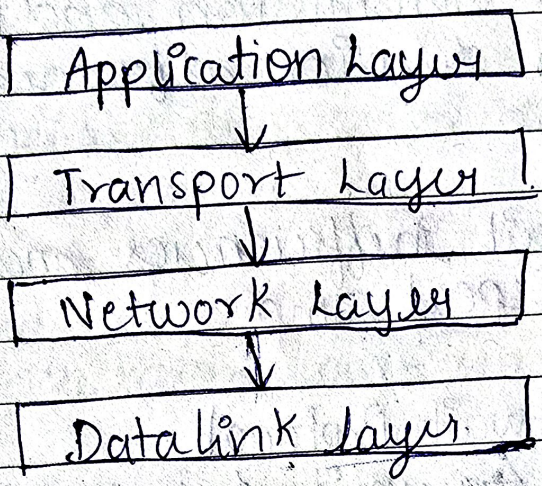
For /24 (mask 255.255.255.0):

Network address : 17.63.110.0
First usable address : 17.63.110.1
Last usable address : 17.63.110.254
Broadcast address : 17.63.110.255

Host range : .1
through : .254;
.0 is network.
.255 is brodcast.

(Q4)

TCP/IP



Application Layer
↳ Procss-to-process communication
↳ Application-specific semantics (SMTP mail delivery rules)

Transport Layer
↳ Reliable byte-stream delivery (TCP)
-ensure in-order delivery
↳ Multiplexing / demultiplexing using ports - allows apps to.

Network Layer
↳ Routing - find path b/w source and destinatio across network.
↳ Logical addressing - IP addressing and fragmentation.

Data link layer
↳ Frame transmission and media access
↳ Error detection at frame-level.

(Q5) Stop & wait inefficiencies and how Selective Repeat Improves

Stop-and-wait Protocol
↳ Sender transmits 1 frame & waits for ACK.
↳ If Round-Trip Time (RTT) is large compared to frame transmission time, channel remains idle most of the time.
↳ Link Utilization (U)

$$U = \frac{\text{Frame transmission Time}}{\text{Frame Transmission time} + RTT}$$

Selective Repeat ARQ
↳ Sender can transmit multiple frames at once (sliding window)
↳ Only erroneous frames are retransmitted. (not the whole sequence)
↳ Increases throughput & utilization, especially in long-delay links.

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(Q6)

Cyclic Redundancy check

↳ CRC computes remainder of polynomial division of data by generator polynomial,

↳ remainder appended as checksum

↳ Receiver divides received bits by same generator, non-zero remainder ⇒ error detected

↳ Practical result:
detects burst errors and many random errors with high probability

(Q7)

Address size: IPv4 = 32 bit,
IPv6 = 128 bit

Header & option: IPv6 has simplified header & extension headers ⇒ faster routing.

Built in feature: IPv6 supports autoconfiguration and eliminates need to NAT, affects addressing & routing.

(Q8) DHCP

↳ DHCP server manages a pool of IP address

↳ when client requests → Server leases an unused address

↳ DORA

Discover → Offer → Request → Acknowledge

↳ Before assignment, DHCP may send an ARP probe to confirm address not already in use

↳ use lease expiration to reclaim addresses.

(Q9)

↳ TCP (connection oriented)

- Reliable, ordered, but higher latency (retransmissions, head-of-line blocking).

↳ UDP (connectionless)

- faster minimal overhead, no reliability.

Recommendation.

- Real-time streaming / gaming / VoIP
  ↳ UDP (because speed > perfect reliability)
- On-demand video (Netflix, youtube)
  ↳ TCP / HTTP.